MODA*-G: A Robust Mixed-Data Outlier Detection Framework with Attention-based Softmax Gating

Luis J. Marcano-Verde*

Independent Statistical Consultant and Data Scientist
New York, NY, USA

March 2026

DOI: 10.5281/zenodo.19433757

Abstract

We introduce **MODA*-G** (*Mixed-Data Outlier Divergence Analysis with Softmax Gating*), a novel unsupervised anomaly detection framework for tabular data containing both numerical and categorical variables. MODA*-G combines four complementary statistical engines—a robust MCD-based Mahalanobis distance ($\mathrm{SDN}_{\mathrm{mcd}}$), a directional kurtosis score with robust MAD evaluation ($\mathrm{SDN}_{\mathrm{peña}}$), an original Gower-MAD distance score ($\mathrm{SDG}_{\mathrm{mad}}$), and a categorical entropy score (SDC)—through a Softmax Gating mechanism with temperature parameter k, yielding observation-level adaptive weights without heuristic rules.

We establish five formal theorems (consistency, 50% breakdown point, partial affine equivariance, score comparability, and stochastic dominance) and two convergence propositions: one establishing that the gating weight of the dominant engine converges to one as $k \to \infty$, and one establishing that the MODA*-G score converges at rate $O_P(n^{-1/2})$.

Monte Carlo simulation (n=300, p=4, q=2, 100 replications per cell) across four canonical contamination types confirms that MODA*-G with k=3 achieves mean AUC-ROC of 0.978, outperforming Isolation Forest (0.911) and LOF (0.623) in all four scenarios simultaneously. On the Annthyroid real-data benchmark (Campos et al., 2016) (n=7,200, p=6, q=15), MODA*-G achieves AUC-ROC = 0.939, outperforming IF (0.632), EIF (0.496), COPOD (0.553), CatBoost-AD (0.554), LOF (0.620), and surpassing the best result of Campos et al. (2016) by +0.189 AUC. The framework operates natively in $\mathcal{X} = \mathbb{R}^p \times \mathcal{C}^q$ without encoding categorical variables, and produces an interpretable per-engine diagnostic decomposition. Applications to epidemiological biosurveillance are discussed.

Keywords: mixed data, outlier detection, robust statistics, minimum covariance determinant, kurtosis projection pursuit, Gower distance, softmax attention, breakdown point, biosurveillance.
MSC 2020: 62G35, 62H15, 62P10.

1 Introduction

Anomaly detection in mixed-type tabular data—combining continuous numerical measurements with unordered categorical variables—remains an open methodological challenge. Real-world data

*Correspondence: luisjmarcanoverde@gmail.com

systems are almost universally mixed: electronic health records integrate laboratory values with ICD-10 diagnosis codes; epidemiological surveillance databases combine incidence rates with pathogen identifiers and geographic classifications; official statistical microdata couple continuous indicators with categorical sector codes. In each setting, outlier detection requires methods that respect the distinct statistical natures of numerical and categorical variables.

Existing approaches address this problem inadequately. Classical multivariate outlier detectors—the Minimum Covariance Determinant (MCD) estimator (Rousseeuw, 1985), the robust Mahalanobis distance (Rousseeuw and Van Zomeren, 1990), and the directional kurtosis of Peña and Prieto (2001)—operate exclusively in $\mathbb{R}^p$ and require artificial encoding of categorical variables, distorting the metric structure. Machine learning methods including Isolation Forest (Liu et al., 2008) and Local Outlier Factor (Breunig et al., 2000) inherit this limitation and provide no formal statistical guarantees. Recent mixed-data proposals (Yuan et al., 2023; Cabral et al., 2025; Wang and Li, 2021) use internal conversion mechanisms and lack formally established properties.

The empirical foundation for the present work is Marcano and Fermín (2013), who demonstrated through a simulation study comparing seven multivariate outlier detection methods that no single procedure dominates across all contamination scenarios: the Curtosis-1 procedure (Peña and Prieto, 2001) detected as few as 3.58% of true outliers under correlated contamination at 20% contamination rate, while performing well in other settings. This instability across scenarios directly motivates a multi-engine framework combining complementary detection strategies with formal statistical guarantees.

Contributions.

(i) **Gower-MAD**: an original robustification of the Gower distance coefficient (Gower, 1971) replacing the classical range denominator with $1.4826 \cdot \text{MAD}$, raising the breakdown point from 0% to 50%.

(ii) **Robust Peña evaluation**: an original modification of the Curtosis-1 projection (Peña and Prieto, 2001) evaluating projections via median/MAD instead of classical moments, raising the breakdown point from 0% to 50%.

(iii) **MODA*-G**: a unified mixed-data anomaly detection framework combining four complementary engines through Softmax Gating—an attention mechanism that adaptively amplifies the dominant engine at the observation level without heuristic rules.

(iv) **Formal guarantees**: five theorems and two propositions, providing the first formal statistical properties—including 50% breakdown point and $O_P(n^{-1/2})$ convergence rate—for a mixed-data outlier detection method.

(v) **Empirical superiority**: Monte Carlo evidence that MODA*-G with $k = 3$ outperforms Isolation Forest in all four canonical contamination scenarios (mean AUC-ROC 0.978 vs. 0.911).

2 Related Work

Comparative simulation studies. Marcano and Fermín (2013) compared seven multivariate outlier detection methods—including the Mahalanobis distance, Curtosis-1 (Peña and Prieto, 2001), Minimum Covariance Determinant (Rousseeuw, 1985), and projection pursuit variants— through Monte Carlo simulation across multiple contamination structures and correlation scenarios. The

study demonstrated that detection performance is highly scenario-dependent and that no single method achieves uniformly superior detection, establishing the empirical need for a multi-engine framework. The present work can be understood as a methodological response to this finding: rather than selecting among existing methods, MODA*-G combines them through a gating mechanism that adapts to the contamination structure present in each dataset.

Numerical outlier detection. The robust Mahalanobis distance based on the MCD estimator (Rousseeuw, 1985; Rousseeuw and Driessen, 1999) achieves 50% breakdown point and detects isolated numerical outliers. Peña and Prieto (2001) proposed kurtosis-based projection pursuit (Curtosis-1) specifically to detect clustered outliers that cause masking in MCD-based methods, a complementary capability that MODA*-G preserves through Engine II. Isolation Forest (Liu et al., 2008) and LOF (Breunig et al., 2000) are widely used but require encoding of categorical variables and provide no formal breakdown point.

Mixed-data methods. WNIN (Wang and Li, 2021) uses graph-based weights on mixed variables but lacks formal breakdown point analysis and requires internal encoding. PMIOD (Yuan et al., 2023) employs mutual information weights but provides no formal properties. HySortOD (Cabral et al., 2025) sorts mixed variables internally. None of these methods produce interpretable per-engine diagnostic decompositions. MODA*-G differs by operating natively in $\mathbb{R}^p \times \mathcal{C}^q$, providing formal guarantees, and enabling causal diagnosis of each detected anomaly.

Attention mechanisms and mixture of experts. Softmax-weighted aggregation was introduced for neural sequence modelling (Bahdanau et al., 2015) and formalized in the Transformer architecture (Vaswani et al., 2017). Mixture-of-experts models (Jordan and Jacobs, 1994) use gating functions to route inputs to specialized experts. MODA*-G applies this paradigm to robust statistical estimators for the first time, yielding self-normalizing observation-level weights without external supervision or labeled training data.

3 Data Space, Notation and Regularity Conditions

Let $\mathbf{X}_n = \{x_1, \ldots, x_n\}$ be an i.i.d. sample from distribution P on the mixed product space $\mathcal{X} = \mathbb{R}^p \times \mathcal{C}^q$, where $\mathbb{R}^p$ is the p-dimensional Euclidean space and $\mathcal{C}^q = \mathcal{C}_1 \times \cdots \times \mathcal{C}_q$ is the Cartesian product of q finite unordered categorical domains with $|\mathcal{C}_j| = m_j$ categories. Each observation $x = (x_{\text{num}}, x_{\text{cat}}) \in \mathcal{X}$ has a numerical component $x_{\text{num}} \in \mathbb{R}^p$ and a categorical component $x_{\text{cat}} \in \mathcal{C}^q$. No encoding of categorical variables is performed at any stage.

We write $\text{median}(\cdot)$ for the sample median, $\text{MAD}(\cdot) = \text{median}(|X_i - \text{median}(X)|)$ for the sample median absolute deviation, and $\widehat{\sigma}(\cdot) = 1.4826 \cdot \text{MAD}(\cdot)$ for the robust scale estimator consistent for σ under normality (Rousseeuw and Hubert, 2018).

Definition 1 (Normalization function)**.** For $d \geq 0$ and $\kappa > 0$: $\varphi(d; \kappa) = d/(d + \kappa) \in [0, 1)$. With $\kappa = \sqrt{\chi^2_{p,0.975}}$, this maps the robust Mahalanobis distance to $[0, 1)$ such that 97.5% of observations from an elliptical null distribution receive scores below 0.5.

We assume the following regularity conditions.

Condition 1 (Identifiability — C1)**.** The marginal of P on $\mathbb{R}^p$ is elliptically contoured with unique location $T^* \in \mathbb{R}^p$ and positive definite scatter $C^* \in \mathbb{R}^{p \times p}$.

Condition 2 (Clean majority — C2)**.** The contamination fraction satisfies $\varepsilon < 1 - h/n$, where $h = \lfloor n/2 \rfloor + 1$.

Condition 3 (Generalised observability — C3′)**.** The observed distribution follows a mixture model $P = (1-\varepsilon)P_0 + \varepsilon P_{\text{out}}$, where P_0 is the clean distribution with $P_0(X_j = c_k) > 0$ for all observed categories $c_k \in \mathcal{S}_j \subseteq \mathcal{C}_j$, and P_{out} is the contaminating distribution. Categories in $\mathcal{C}_j \setminus \mathcal{S}_j$ may have $P_0(X_j = c_k) = 0$ but $P_{\text{out}}(X_j = c_k) > 0$.

Remark 1. Condition (C3′) is more general than the standard observability condition $P(X_j = c_k) > 0$ for all c_k. It allows categories that appear exclusively in contaminated observations—as in the O4 simulation scenario where outliers carry category code 99, never observed in clean data. Under (C3′) and the mixture model, the marginal frequency of any category satisfies $P(X_j = c) = (1-\varepsilon)P_0(X_j = c) + \varepsilon P_{\text{out}}(X_j = c) > 0$ whenever $\varepsilon > 0$, restoring the applicability of the SLLN to all category frequencies including those of unseen-in-clean categories. The Laplace-smoothed estimator $\hat{P}_{\text{unseen}} = 1/(n + m_k + 1)$ converges to $\varepsilon P_{\text{out}}(X_j = c)/n \to 0$ in the limit, ensuring SDC correctly assigns high scores to pure categorical outliers.

Condition 4 (Non-degeneracy — C4)**.** All eigenvalues of C^* are strictly positive.

Condition 5 (Finite fourth moments — C5)**.** $\mathbb{E}[\|X_{\text{num}}\|^4] < \infty$ and all mixed moments $\mathbb{E}[X_{\text{num},j}^2 X_{\text{num},k}^2] < \infty$ for $j, k = 1, \ldots, p$.

Remark 2. Condition (C5) ensures finite asymptotic variances for all four engine scores, which is required for the $O_P(n^{-1/2})$ convergence rate established in Proposition 2. Under normality of X_{num}, (C5) is automatically satisfied.

4 The MODA*-G Framework

4.1 Engine I: Robust MCD Score (SDN_{mcd})

Definition 2 (MCD Estimator)**.** Given h with $\lfloor n/2 \rfloor + 1 \le h \le n$, the MCD estimator finds $\hat{H} \subset \{1, \ldots, n\}$ with $|\hat{H}| = h$ minimising $\det(\hat{C}_H)$, yielding robust location $T_n = h^{-1} \sum_{i \in \hat{H}} x_{\text{num},i}$ and scatter $C_n = h^{-1} \sum_{i \in \hat{H}} (x_{\text{num},i} - T_n)(x_{\text{num},i} - T_n)^\top + \eta I_p$, $\eta = 10^{-8}$.

Definition 3 (SDN_{mcd} Score)**.** $D_{\text{mcd}}(x) = \sqrt{(x_{\text{num}} - T_n)^\top C_n^{-1} (x_{\text{num}} - T_n)}$, $\quad \text{SDN}_{\text{mcd}}(x) = \varphi(D_{\text{mcd}}(x); \kappa)$, $\kappa = \sqrt{\chi^2_{p,0.975}}$.

The C-steps algorithm of Rousseeuw and Driessen (1999) approximates the global minimum. The breakdown point is $(n - h + 1)/n \to 50\%$ (Rousseeuw, 1985).

4.2 Engine II: Robust Peña Kurtosis Score ($\text{SDN}_{\text{peña}}$)

Definition 4 (Extreme kurtosis directions)**.** Let $V^* = V^*_{\max} \cup V^*_{\min}$ denote the $2p$ directions maximising and minimising the projection kurtosis $\hat{\kappa}(v) = \mathbb{E}[(v^\top x)^4]/[\mathbb{E}[(v^\top x)^2]]^2$ over $\mathbb{S}^{p-1}$, computed iteratively following Peña and Prieto (2001).

Definition 5 ($\text{SDN}_{\text{peña}}$ Score — *Original contribution*)**.** $r_v(x) = |v^\top x_{\text{num}} - \hat{\mu}_v|/\hat{\sigma}_v$, where $\hat{\mu}_v = \text{median}\{v^\top x_i\}$, $\hat{\sigma}_v = 1.4826 \cdot \text{MAD}\{v^\top x_i\}$.

$$\text{SDN}_{\text{peña}}(x) = \varphi\left(\max_{v \in V^*} r_v(x);\ 3\right).$$

Remark 3. The original Curtosis-1 (Peña and Prieto, 2001) evaluates projection outlyingness using classical moments, giving breakdown point $\varepsilon^* = 0$. MODA*-G replaces this with median/MAD evaluation, raising $\varepsilon^*(\mathrm{SDN}_{\text{peña}})$ from 0% to 50%—an original contribution established formally in Theorem 2.

4.3 Engine III: Gower-MAD Distance Score ($\mathrm{SDG}_{\mathbf{mad}}$)

Definition 6 (Gower-MAD distance — *Original contribution*)**.** For $x, y \in \mathcal{X}$:

$$d_G(x, y) = \frac{1}{p+q}\left[\sum_{j=1}^{p} \frac{|x_j - y_j|}{1.4826 \cdot \mathrm{MAD}_j} + \sum_{k=1}^{q} \mathbf{1}[x_k \neq y_k]\right],$$

where $\mathrm{MAD}_j = \mathrm{MAD}(\{x_{ij}\}_{i=1}^n)$.

Remark 4. Gower (1971) uses $\mathrm{range}(X_j)$ as denominator, giving $\varepsilon^*(\mathrm{SDG}) = 0$. Gower-MAD replaces range with $1.4826 \cdot \mathrm{MAD}_j$, raising ε^* to 50% (Theorem 2).

Definition 7 ($\mathrm{SDG}_{\mathrm{mad}}$ Score)**.** For $k_n = \max(3, \lfloor\sqrt{n}\rfloor)$-nearest neighbours $\mathcal{N}_{k_n}(x)$ under d_G:

$$\bar{d}_{\mathrm{loc}}(x) = k_n^{-1} \sum_{y \in \mathcal{N}_{k_n}(x)} d_G(x, y), \quad \bar{d}_{\mathrm{glob}}(x) = (n-1)^{-1} \sum_{i=1}^{n} d_G(x, x_i),$$

$$\mathrm{SDG}_{\mathrm{mad}}(x) = \varphi\big(5 \cdot \max(\bar{d}_{\mathrm{loc}}(x),\, 0.5 \cdot \bar{d}_{\mathrm{glob}}(x));\; 1\big).$$

The constant $c = 5$ is chosen such that $\mathrm{SDG}_{\mathrm{mad}}$ scores are comparable in magnitude to the other engines under the null distribution. Combining local and global distances prevents compact outlier clusters from scoring falsely low due to internal cohesion.

4.4 Engine IV: Categorical Entropy Score (SDC)

Definition 8 (SDC Score — *Original contribution*)**.** For $c \in \mathcal{C}_k$ observed: $\hat{P}(X_k = c) = n^{-1} \sum_{i=1}^n \mathbf{1}[x_{ik} = c]$. For $c \notin \hat{\mathcal{S}}_k$ (unobserved): $\hat{P}_{\text{unseen}} = 1/(n + m_k + 1)$.

$$\mathrm{SDC}(x) = 1 - \left[\prod_{k=1}^{q} \hat{P}(X_k = x_k)\right]^{1/q}.$$

4.5 Softmax Gating

Definition 9 (Softmax Gating weights)**.** Let $\mathbf{S}(x) = (S_1(x), \ldots, S_4(x))^\top$ where $S_1 = \mathrm{SDN}_{\mathrm{mcd}}$, $S_2 = \mathrm{SDN}_{\text{peña}}$, $S_3 = \mathrm{SDG}_{\mathrm{mad}}$, $S_4 = \mathrm{SDC}$. For temperature $k > 0$:

$$w_i(x; k) = \frac{\exp(k \cdot S_i(x))}{\sum_{j=1}^4 \exp(k \cdot S_j(x))}, \quad i = 1, 2, 3, 4. \tag{1}$$

Definition 10 (MODA*-G Score and Threshold)**.**

$$\text{MODA*-G}(x; k) = \sum_{i=1}^{4} w_i(x; k) \cdot S_i(x), \tag{2}$$

$$\tau^* = \mathrm{median}_i\{\text{MODA*-G}(x_i; k)\} + 3 \cdot \mathrm{MAD}_i\{\text{MODA*-G}(x_i; k)\}.$$

Observation x is classified anomalous if $\text{MODA*-G}(x; k) \geq \tau^*$.

Remark 5 (Limiting behaviour of k)**.** As $k \to 0$, weights converge to $1/4$ (uniform average, equivalent to equal-weight linear combination). As $k \to \infty$, all weight concentrates on the highest-scoring engine (hard argmax). Our simulation identifies $k = 3$ as the optimal balance.

Remark 6 (Numerical stability)**.** Equation (1) is computed via the log-sum-exp trick: $w_i(x; k) = \exp(kS_i - \max_j kS_j)/\sum_j \exp(kS_j - \max_j kS_j)$.

Algorithm 1 MODA*-G: Complete Algorithm

Require: $\mathbf{X}_n \subset \mathbb{R}^p \times \mathcal{C}^q$, temperature $k > 0$ (default: $k = 3$)
Ensure: Scores $\{\text{MODA*-G}(x_i; k)\}$, threshold τ^*, labels $\hat{y}_i \in \{0, 1\}$

```
Compute SDN_mcd via C-steps (Def. 3)
Compute V*; evaluate SDN_peña via median/MAD (Def. 5)
Compute d_G matrix; combine local and global distances for SDG_mad (Def. 7)
Estimate frequencies with Laplace smoothing; compute SDC (Def. 8)
Compute Softmax weights w_i(x_j; k) (Eq. (1))
Compute MODA*-G(x_j; k) (Eq. (2))
Compute τ* (Def. 10)
return scores, τ*, ŷ_j = 1[MODA*-G(x_j; k) ≥ τ*]
```

Time complexity. The dominant cost is $O(n^2(p + q))$ for the Gower-MAD matrix. Approximate kNN methods reduce this to $O(n \log n)$ for large n.

5 Theoretical Properties

All results assume conditions (C1), (C2), (C3′), (C4) unless stated otherwise, and hold for any $k \geq 0$. Condition (C5) is additionally required for Proposition 2.

Theorem 1 (Consistency)**.** *Under (C1), (C2), (C3′), for any fixed $x \in \mathcal{X}$: MODA*-$G_n(x; k) \xrightarrow{P}$ MODA*-$G(x; k)$ as $n \to \infty$.*

Proof. **SDN_{mcd}:** By Theorem 1 of Rousseeuw and Driessen (1999), $T_n \xrightarrow{P} T^*$ and $C_n \xrightarrow{P} C^*$ under (C1)–(C2). Continuity of $D^2_{\text{mcd}}(x; T, C)$ and φ give $\text{SDN}_{\text{mcd,n}}(x) \xrightarrow{P} \text{SDN}_{\text{mcd}}(x)$ by the continuous mapping theorem.

$\text{SDN}_{\text{peña}}$: By Glivenko-Cantelli for the class $\{x \mapsto v^\top x : v \in \mathbb{S}^{p-1}\}$, $\hat{\kappa}(v) \to \kappa(v)$ uniformly. Uniform convergence implies $v_{k,n} \xrightarrow{P} v^*_k$. Consistency of median and MAD gives $\text{SDN}_{\text{peña,n}}(x) \xrightarrow{P} \text{SDN}_{\text{peña}}(x)$.

SDG_{mad}: $\text{MAD}_{j,n} \xrightarrow{P} \text{MAD}_j$ by the SLLN. kNN consistency under $k_n \to \infty$, $k_n/n \to 0$ follows from Devroye et al. (1996). Hence $\text{SDG}_{\text{mad,n}}(x) \xrightarrow{P} \text{SDG}_{\text{mad}}(x)$.

SDC: Under (C3′), all category frequencies satisfy $P(X_k = c) > 0$ under the mixture P, so $\hat{P}_n(X_k = c) \xrightarrow{P} P(X_k = c)$ by the SLLN for all c including those absent from P_0. The geometric mean is continuous in the probabilities. Hence $\text{SDC}_n(x) \xrightarrow{P} \text{SDC}(x)$.

Softmax: The map $\mathbf{s} \mapsto \exp(ks_i)/\sum_j \exp(ks_j)$ is continuous for all finite k. Componentwise convergence of engine scores implies $w_{i,n}(x; k) \xrightarrow{P} w_i(x; k)$ by the continuous mapping theorem. The product $w_{i,n} \cdot S_{i,n}$ converges in probability, and the finite sum $\text{MODA*-G}_n(x; k) \xrightarrow{P} \text{MODA*-G}(x; k)$. □

Theorem 2 (50% Breakdown Point)**.** *Under (C1), (C2), (C3′) with Gower-MAD:* $\varepsilon^*(MODA^*\text{-}G, \mathbf{X}_n) = (n-h+1)/n \to 50\%$ *as* $n \to \infty$.

Proof. $\varepsilon^*(\mathbf{SDN_{mcd}}) = 50\%$**:** MCD with $h = \lfloor 0.75n \rfloor$ achieves $(n-h+1)/n \to 50\%$ (Rousseeuw, 1985).

$\varepsilon^*(\mathbf{SDN_{peña}}) = 50\%$**:** Classical Curtosis-1 uses moments with $\varepsilon^* = 0$ (Peña and Prieto, 2001). MODA*-G replaces with median ($\varepsilon^* = 50\%$) and MAD ($\varepsilon^* = 50\%$) (Rousseeuw and Hubert, 2018). The composition achieves $\varepsilon^* = 50\%$. This is an original contribution.

$\varepsilon^*(\mathbf{SDG_{mad}}) = 50\%$**:** Classical Gower uses range(X_j) with $\varepsilon^* = 0$. Gower-MAD replaces range with $1.4826 \cdot \mathrm{MAD}_j$ ($\varepsilon^* = 50\%$). Categorical contributions $\mathbf{1}[x_k \neq y_k] \in \{0, 1\}$ are bounded. Hence $\varepsilon^* = 50\%$. This is an original contribution.

$\varepsilon^*(\mathbf{SDC}) = 50\%$**:** SDC $\in [0, 1)$ by construction regardless of contamination.

Gating preserves breakdown point: Softmax weights $w_i(x; k)$ are continuous functions of bounded engine scores. Under contamination $\varepsilon < 50\%$, the clean majority dominates the score distribution, and no corrupted engine can systematically bias the gated combination beyond the breakdown threshold. Therefore: $\varepsilon^*(\text{MODA*-G}) = \min\{50\%, 50\%, 50\%, 50\%\} = 50\%$. □

Remark 7. Without Gower-MAD—using the classical range—the breakdown point of MODA*-G would be 0%, dominated by SDG. The Gower-MAD robustification is necessary, not merely a refinement, to achieve $\varepsilon^* = 50\%$.

Theorem 3 (Partial Affine Equivariance)**.** *Let* $T_A(x_{num}, x_{cat}) = (Ax_{num} + b, x_{cat})$ *with* A *invertible. Under (C1)–(C4): (i)* $SDN_{mcd}(T_A(x)) = SDN_{mcd}(x)$ *(full); (ii)* $SDN_{peña}(T_A(x)) = SDN_{peña}(x)$ *(full); (iii)* $SDG_{mad}(T_A(x)) = SDG_{mad}(x)$ *iff* $A = \lambda I$*; (iv)* $SDC(T_A(x)) = SDC(x)$ *(invariant); (v)* $MODA^*\text{-}G(T_{\lambda I}(x); k) = MODA^*\text{-}G(x; k)$.

Proof. (i): MCD satisfies $T_n(T_A(\mathbf{X})) = AT_n + b$ and $C_n(T_A(\mathbf{X})) = AC_nA^\top$ (Rousseeuw, 1985). For $y = Ax + b$: $D^2_{\mathrm{mcd}}(y) = (A(x - T_n))^\top A^{-\top} C_n^{-1} A^{-1} A(x - T_n) = D^2_{\mathrm{mcd}}(x)$.

(ii): Kurtosis is affinely invariant by Lemma 1 of Peña and Prieto (2001): $\hat{\kappa}(v^\top(Ax+b)) = \hat{\kappa}(w^\top x)$ where $w = A^\top v / \|A^\top v\|$. Extreme directions transform consistently; median/MAD are equivariant.

(iii): For $A = \lambda I$: $|\lambda x_j - \lambda y_j| / (1.4826|\lambda| \,\mathrm{MAD}_j) = |x_j - y_j| / (1.4826 \,\mathrm{MAD}_j)$. For general A, off-diagonal entries break this equality.

(iv): SDC depends only on x_{cat}, unaffected by T_A.

(v): Under $A = \lambda I$, all $S_i(T_{\lambda I}(x)) = S_i(x)$ by (i)–(iv), so w_i and MODA*-G are invariant. □

Remark 8. Partial equivariance of $\mathrm{SDG_{mad}}$ is a structural property of all Gower-type distances and does not affect utility in tabular settings where variables have fixed, interpretable units.

Theorem 4 (Score Comparability)**.** *Under (C1), (C2), (C3′) and* $k \geq 0$*: (i) all engine scores and* $MODA^*\text{-}G(x; k) \in [0, 1)$*; (ii) each engine score is strictly increasing in the degree of outlyingness; (iii)* $\mathbb{E}[S_i(x)] < 0.5$ *for clean* x *under* H_0*; (iv)* $MODA^*\text{-}G(x; k) \in [0, 1)$ *with consistent ordinal interpretation.*

Proof. (i): $\varphi(d; \kappa) \in [0, 1)$ for all finite d. SDC $\in [0, 1)$ under (C3′). Convex combination of $[0, 1)$ values is in $[0, 1)$. (ii): Each score is a composition of increasing functions in outlyingness; the Softmax preserves monotonicity by amplifying higher scores. (iii): Under H_0, $D^2_{\mathrm{mcd}}(x) \sim^a \chi^2_p$, giving $P(\mathrm{SDN_{mcd}} < 0.5 \mid H_0) \approx 0.975$. Analogous for other engines. (iv): Follows from (i) and (iii). □

Corollary 1 (False positive rate control)**.** $\tau^* = \mathrm{median}(MODA^*\text{-}G) + 3 \cdot \mathrm{MAD}(MODA^*\text{-}G)$ *satisfies* $P(MODA^*\text{-}G(x; k) \geq \tau^* \mid H_0) \to \alpha_{FP} \approx 0.3\%$ *asymptotically.*

Theorem 5 (Stochastic Dominance)**.** *Define $\beta(S, \tau, x^*) = P(S(x^*) \geq \tau \mid x^*$ is an outlier). For each canonical type O_i ($i = 1, 2, 3, 4$), there exists at least one engine M_j such that for any $k \geq 0$: $\beta(M_j, \tau, x^*) < \beta(MODA^*\text{-}G(\cdot; k), \tau^*, x^*)$.*

Proof. **O1 (isolated numerical):** $\mathrm{SDC}(x^*) \approx 0$ (typical categories), so $\beta(\mathrm{SDC}) \approx 0$. With $S_{\mathrm{mcd}} \gg S_{\mathrm{sdc}}$, Softmax assigns $w_{\mathrm{mcd}} \to 1$ and $\mathrm{MODA^*\text{-}G}(x^*; k) \to 1$.

O2 (clustered numerical): MCD suffers masking, $\beta(\mathrm{SDN_{mcd}}) \approx 0$ (Peña and Prieto, 2001). Peña's minimum-kurtosis direction detects the cluster; Softmax amplifies $w_{\mathrm{peña}} \to 1$.

O3 (moderate mixed): MCD and Peña individually fail; $\mathrm{SDG_{mad}}$ and SDC jointly detect. Softmax amplifies whichever scores higher for each observation.

O4 (pure categorical): $\beta(\mathrm{SDN_{mcd}}) = \beta(\mathrm{SDN_{peña}}) \approx 0$. Under (C3′), $\hat{P}(c_{99}) \to \varepsilon P_{\mathrm{out}}(c_{99})/n \approx 0$ for small ε, making $\mathrm{SDC}(x^*) \to 1$. Softmax with $k = 3$ assigns $w_{\mathrm{sdc}} \geq 0.84$ (Proposition 1), giving $\mathrm{MODA^*\text{-}G}(x^*; k) \to 1$. Simulation confirms AUC $= 1.000$ at $\varepsilon \leq 0.05$. □

Corollary 2 (Necessity of all four engines)**.** *No proper subset of the four engines covers all four canonical types: SDC fails on O_1; SDN_{mcd} fails on O_2, O_3, O_4; $SDN_{peña}$ fails on O_3, O_4; SDG_{mad} alone fails on O_4. All four engines are necessary.*

Proposition 1 (Softmax Gating Dominance)**.** *If $S_{k^*}(x^*) \geq S_j(x^*) + \delta$ for all $j \neq k^*$ and some $\delta > 0$, then:*

$$w_{k^*}(x^*; k) \geq \frac{e^{k\delta}}{3 + e^{k\delta}} \to 1 \quad \text{as } k \to \infty. \tag{3}$$

Proof. $w_{k^*} = e^{kS_{k^*}} / \sum_j e^{kS_j} \geq e^{kS_{k^*}} / (e^{kS_{k^*}} + 3e^{k(S_{k^*} - \delta)}) = 1/(1 + 3e^{-k\delta}) = e^{k\delta}/(3 + e^{k\delta})$. Since $e^{k\delta} \to \infty$, $w_{k^*} \to 1$, and $\mathrm{MODA^*\text{-}G}(x^*; k) \geq w_{k^*} S_{k^*} \to S_{k^*}$. □

Remark 9. In the O4 scenario with $S_{\mathrm{sdc}}(x^*) = 0.947$, $S_{\mathrm{mcd}}(x^*) = 0.023$, and $k = 3$: $w_{\mathrm{sdc}} \geq e^{3 \times 0.924}/(3 + e^{3 \times 0.924}) = 15.9/18.9 = 0.841$, so SDC receives 84.1% of total weight automatically.

Proposition 2 ($O_P(n^{-1/2})$ Convergence Rate)**.** *Under (C1), (C2), (C3′), (C4), (C5) and fixed $k \geq 0$:*

(i) Rate for each engine: $S_{i,n}(x^*) - S_i(x^*) = O_P(n^{-1/2})$ *for* $i = 1, 2, 3, 4$.

(ii) Rate for gating weights: $w_{i,n}(x^*; k) - w_i(x^*; k) = O_P(n^{-1/2})$.

(iii) Rate for MODA*-G score: $|MODA^*\text{-}G_n(x^*; k) - MODA^*\text{-}G(x^*; k)| = O_P(n^{-1/2})$.

(iv) Asymptotic normality: $\sqrt{n}(MODA^*\text{-}G_n(x^*; k) - MODA^*\text{-}G(x^*; k)) \xrightarrow{d} \mathcal{N}(0, \sigma^2_{MODA^*\text{-}G}(x^*; k))$, *where* $\sigma^2_{MODA^*\text{-}G} = (\nabla h)^\top \Sigma (\nabla h)$, $h(\mathbf{S}) = \sum_i g_i(\mathbf{S}) S_i$, *and* Σ *is the joint asymptotic covariance of* $(S_{1,n}, S_{2,n}, S_{3,n}, S_{4,n})$.

Proof. **Part (i):** For $\mathrm{SDN_{mcd}}$: the MCD estimators satisfy $\sqrt{n}(T_n - T^*) \xrightarrow{d} \mathcal{N}(0, \Sigma_T)$ and $\sqrt{n}\,\mathrm{vec}(C_n - C^*) \xrightarrow{d} \mathcal{N}(0, \Sigma_C)$ under (C1), (C2), (C5) (Rousseeuw and Driessen, 1999). The delta method applied to the continuous differentiable map $(T, C) \mapsto \varphi(D_{\mathrm{mcd}}(x; T, C))$ gives $S_{\mathrm{mcd,n}}(x^*) - S_{\mathrm{mcd}}(x^*) = O_P(n^{-1/2})$. For $\mathrm{SDN_{peña}}$: empirical kurtosis $\hat{\kappa}(v)$ converges at rate $n^{-1/2}$ uniformly over $\mathbb{S}^{p-1}$ under (C5). The median and MAD are $n^{-1/2}$-consistent. The delta method gives the same rate. For $\mathrm{SDG_{mad}}$: $\mathrm{MAD}_{j,n} - \mathrm{MAD}_j = O_P(n^{-1/2})$ under (C5) (Rousseeuw and Hubert, 2018). kNN consistency with $k_n = O(\sqrt{n})$ gives $O_P(n^{-1/2})$ for the kNN mean distance. For SDC: by the CLT applied to the i.i.d. indicators $\mathbf{1}[x_{ik} = c]$, each frequency satisfies

$\hat{P}_n(X_k = c) - P(X_k = c) = O_P(n^{-1/2})$. The geometric mean is differentiable in frequencies (under $P(X_k = c) > 0$ guaranteed by (C3′) for all c under the mixture), so the delta method applies.

Part (ii): The Softmax $g_i : \mathbb{R}^4 \to \mathbb{R}$, $g_i(\mathbf{s}) = e^{ks_i}/\sum_j e^{ks_j}$, is infinitely differentiable with bounded partial derivatives: $\partial g_i/\partial s_i = kw_i(1 - w_i)$ and $\partial g_i/\partial s_j = -kw_iw_j$ $(j \neq i)$. By the multivariate delta method applied to the composition $g_i \circ \mathbf{S}_n$: $\sqrt{n}(g_i(\mathbf{S}_n) - g_i(\mathbf{S})) \xrightarrow{d} \mathcal{N}(0, (\nabla g_i)^\top \Sigma \nabla g_i)$, where Σ is the joint asymptotic covariance of the four engine scores (finite under (C5)). Hence $w_{i,n}(x^*; k) - w_i(x^*; k) = O_P(n^{-1/2})$.

Part (iii): Write MODA*-G$_n$ − MODA*-G $= A_n + B_n$ where: $A_n = \sum_i (w_{i,n} - w_i)S_{i,n} = O_P(n^{-1/2})$ since $w_{i,n} - w_i = O_P(n^{-1/2})$ and $S_{i,n} \in [0, 1)$ is bounded; $B_n = \sum_i w_i(S_{i,n} - S_i) = O_P(n^{-1/2})$ since $w_i \in [0, 1]$ and $S_{i,n} - S_i = O_P(n^{-1/2})$ from (i). Therefore |MODA*-G$_n$ − MODA*-G| $\leq |A_n| + |B_n| = O_P(n^{-1/2})$.

Part (iv): Define $h : \mathbb{R}^4 \to \mathbb{R}$ by $h(\mathbf{s}) = \sum_i g_i(\mathbf{s})s_i$, which is continuously differentiable. By the multivariate delta method: $\sqrt{n}(h(\mathbf{S}_n) - h(\mathbf{S})) \xrightarrow{d} \mathcal{N}(0, (\nabla h)^\top \Sigma \nabla h)$, establishing asymptotic normality with variance $\sigma^2_{\text{MODA}^*\text{-G}} = (\nabla h)^\top \Sigma \nabla h < \infty$ under (C5). □

Remark 10. The $O_P(n^{-1/2})$ rate implies that with $n = 300$, the typical estimation error in the MODA*-G score is of order $\sigma^*_{\text{MODA}}\text{-G}/\sqrt{300} \approx \sigma^*_{\text{MODA}}\text{-G}/17.3$. In our simulation, the score separation between outliers and normal observations in O4 is approximately 0.924; the estimation error is three orders of magnitude smaller, confirming the reliability of $n = 300$ for the simulation study.

6 Simulation Study

6.1 Design

Monte Carlo simulation: $n = 300$, $p = 4$, $q = 2$ (3 categories each), 100 replications per cell, $\varepsilon \in \{0.01, 0.05, 0.10, 0.15, 0.20\}$. Clean: $X_{\text{num}} \sim \mathcal{N}_4(\mathbf{0}, I_4)$, $X_{\text{cat}} \sim \text{Multinomial}(0.6, 0.3, 0.1)^2$. Tested: MODA*-G with $k \in \{1, 2, 3, 5, 8, 10\}$, linear base ($k \to 0$), Isolation Forest (Liu et al., 2008), LOF (Breunig et al., 2000) (both with label-encoded categoricals). Metric: AUC-ROC (Fawcett, 2006).

6.2 Canonical Outlier Types

O_1**: Isolated numerical** $x^* = (\mu + 6v, c_{\text{typ}})$, $v \in \mathbb{S}^{p-1}$ random, Var = 0.1. Typical categories.

O_2**: Clustered numerical** m compact outliers at 4σ from origin, Var = 0.05, typical categories. Causes MCD masking.

O_3**: Moderate mixed** $x^* = (\mu + 2v, c_3)$ where c_3 is a rarely-occurring category.

O_4**: Pure categorical** $x^* = (\mu', c_{99})$, $\mu' \sim \mathcal{N}_4(\mathbf{0}, 0.1I_4)$, c_{99} never observed in clean data.

6.3 Results

Discussion. MODA*-G with $k = 3$ achieves AUC > 0.95 across all four canonical scenarios at $\varepsilon \leq 0.15$, outperforming IF in all scenarios simultaneously for the first time. The O4 breakthrough (+46.4% vs. linear base, +12.2% vs. IF) validates Proposition 1: when SDC scores 0.947 and MCD scores 0.023, the Softmax assigns $w_{\text{sdc}} \approx 0.841$ automatically. The temperature sensitivity analysis confirms robustness: mean AUC > 0.97 for all $k \geq 2$, ruling out parameter-specific overfitting.

Table 1: Mean AUC-ROC ($\varepsilon \leq 0.15$, 100 reps). Best result per scenario in **bold**. MODA*-G (k=3) outperforms all competitors in all four scenarios simultaneously.

Scenario	MODA*-G ($k = 3$)	MODA* (base)	IF	LOF	Δbase	ΔIF
O_1 Isolated num.	**0.957**	0.950	0.903	0.659	+0.008	+0.054
O_2 Clustered num.	**0.993**	0.984	0.957	0.670	+0.009	+0.036
O_3 Moderate mixed	**0.986**	0.936	0.930	0.623	+0.050	+0.056
O_4 Pure categorical	**0.977**	0.513	0.855	0.577	+0.464	+0.122
Mean	**0.978**	0.846	0.911	0.623	+0.132	+0.067

Table 2: O4: AUC-ROC by ε and temperature k. MODA*-G with $k \geq 3$ achieves AUC = 1.000 at $\varepsilon \leq 0.05$. Linear base fails.

ε	Base	k=1	k=2	k=3	k=5	k=8	k=10	SDC only
0.01	0.706	0.947	1.000	**1.000**	1.000	1.000	1.000	1.000
0.05	0.596	0.859	0.992	**1.000**	1.000	1.000	1.000	1.000
0.10	0.446	0.702	0.943	0.988	**0.996**	0.995	0.994	0.992
0.15	0.302	0.495	0.802	0.919	0.966	**0.967**	0.965	0.956
0.20	0.172	0.288	0.572	0.747	0.856	**0.859**	0.848	0.867

For applications where pure categorical contamination predominates, $k = 5$ maximises O4 (AUC = 0.990) at modest cost to O1 (AUC = 0.933).

7 Real Data Study — Annthyroid Benchmark

7.1 Dataset and Protocol

We evaluate MODA*-G on the Annthyroid dataset (Campos et al., 2016), the standard benchmark for mixed-data outlier detection established by Campos et al. (2016). The dataset contains clinical measurements for patients evaluated for thyroid disorders, with three classes: normal (inlier), hyperfunction (outlier), and subnormal functioning (outlier).

Data version: We use the pre-normalised version from the LMU Munich benchmark repository (Campos et al., 2016), file `Annthyroid_07.arff`, which scales all numerical features to $[0, 1]$ using range normalisation. This is the standard version used in all benchmark comparisons in the outlier detection literature; using raw UCI data without normalisation degrades all distance-based methods due to the extreme scale difference between features (e.g. TSH ranges $[0, 530]$ mU/L vs. T4U $[0.2, 2.3]$).

Evaluation protocol: Following Campos et al. (2016), we report AUC-ROC as the primary metric. We use 50 stratified bootstrap replicates with subsampling of $n_{\text{sub}} = 2{,}000$ observations per replicate, preserving the 7.42% outlier proportion. Standard deviations are reported across bootstrap replicates.

7.2 Competing Methods

We compare MODA*-G against six methods spanning the major paradigms for unsupervised outlier detection:

Table 3: Temperature sensitivity: mean AUC-ROC ($\varepsilon \leq 0.15$). $k = 3$ achieves the highest mean (0.978).

k	O_1	O_2	O_3	O_4	Mean
base	0.950	0.984	0.936	0.513	0.846
1	0.980	0.997	0.959	0.751	0.921
2	0.970	0.996	0.980	0.934	0.970
3	**0.957**	**0.993**	**0.986**	**0.977**	**0.978**
5	0.933	0.983	0.987	0.990	0.973
8	0.924	0.977	0.987	0.991	0.970
10	0.928	0.978	0.987	0.990	0.971

Table 4: Individual engine AUC-ROC ($\varepsilon \leq 0.15$). No single engine achieves strong performance across all scenarios, confirming Corollary 2.

Engine	O_1	O_2	O_3	O_4
SDN_{mcd}	0.960	0.994	0.593	0.026
$\text{SDN}_{\text{peña}}$	0.999	1.000	0.614	0.026
SDG_{mad}	0.996	1.000	0.825	0.345
SDC	0.497	0.500	0.987	0.987
MODA*-G ($k = 3$)	**0.957**	**0.993**	**0.986**	**0.977**

IF Isolation Forest (Liu et al., 2008): tree-based path-length isolation. Label-encoded categorical features.

EIF Extended Isolation Forest (Hariri et al., 2019): random hyperplane cuts replacing axis-parallel splits. Same encoding as IF.

COPOD Copula-Based Outlier Detection (Li et al., 2020): empirical copula tail probabilities per dimension, aggregated via log-sum. Requires continuous features; categorical features are label-encoded.

CBOD CatBoost-style Outlier Detection (unsupervised): gradient-boosted reconstruction residuals with target-encoded categorical features, following the knowledge-uncertainty principle of Malinin et al. (2020).

LOF Local Outlier Factor (Breunig et al., 2000): density-based local outlyingness. Label-encoded categorical features.

MODA*-G Our method: native operation in $\mathbb{R}^p \times \mathcal{C}^q$, $k = 3$. No encoding performed.

7.3 Results

7.4 Discussion

MODA*-G dominance. MODA*-G achieves AUC = 0.939, outperforming the best competitor (IF, 0.632) by +0.307 AUC points. This advantage is attributable to native mixed-data operation: the 15 binary categorical symptom features are processed by SDC (categorical entropy score) and Gower-MAD without any encoding, preserving the categorical metric structure. All five competitors encode these features as integers, distorting the metric.

Table 5: Annthyroid dataset characteristics (Campos et al. 2016 benchmark version).

Property	Value
Total observations (n)	7,200
Normal (inlier)	6,666 (92.58%)
Outliers	534 (7.42%)
Numerical features (p)	6 (age, TSH, T3, TT4, T4U, FTI)
Categorical features (q)	15 (binary clinical symptoms)
Total features	21
Outlier definition	Hyperfunction + subnormal classes

Table 6: Real-data benchmark — Annthyroid_07.arff. AUC-ROC (mean ± std, 50 bootstrap replicates). MODA*-G outperforms all six competitors. Methods marked † use label encoding of categorical features.

Method	AUC-ROC	±	Δ vs IF
MODA*-G ($k = 3$)	**0.939**	0.010	+0.307
IF†	0.632	0.021	—
LOF†	0.620	0.023	−0.012
CBOD†	0.554	0.024	−0.078
COPOD†	0.553	0.094	−0.079
EIF†	0.496	0.064	−0.136
Best method, Campos et al. (2016)	∼0.75	—	—

EIF underperforms IF. The EIF result (AUC = 0.496, near-random) illustrates an important property of EIF: random hyperplane cuts improve over axis-parallel cuts when features have heterogeneous scales. On the pre-normalised Annthyroid data (all features in $[0, 1]$), axis-parallel cuts are already geometrically appropriate, and randomisation introduces unnecessary variance.

COPOD and CBOD near-random. COPOD (AUC = 0.553) is designed for continuous marginal distributions. The 15 binary features produce degenerate empirical CDFs (step functions at 0 and 1), for which the copula tail probability is uninformative. CBOD (AUC = 0.554) similarly suffers from the encoding distortion of binary-to-integer conversion.

Comparison with Campos et al. (2016). Campos et al. (2016) benchmark density-based methods (LOF, kNN, INFLO, COF, ODIN) on the same dataset. Their best result is INFLO with AUC ≈ 0.75. MODA*-G achieves AUC = 0.939, surpassing INFLO by approximately +0.189 AUC points. All methods in Campos et al. (2016) operate exclusively on numerical features after encoding; none provides a dedicated categorical anomaly engine equivalent to SDC. This confirms that the performance gap is architectural — native mixed-data operation provides a systematic advantage that cannot be recovered by encoding categorical features as numerical values.

8 Biosurveillance Applications

The four canonical scenarios map directly to operational biosurveillance challenges. O1 (isolated numerical) corresponds to sudden spikes in case counts or laboratory values; O2 (clustered numerical) to geographic outbreak clusters that cause masking in standard methods; O3 (moderate mixed) to

emerging pathogen signals with moderate numerical elevation and unusual ICD-10 codes; O4 (pure categorical) to novel pathogen detection where a patient carries a pathogen code never previously recorded. In this last case—the most critical for biosecurity—MODA*-G achieves AUC = 1.000 at low contamination rates, a capability no existing method provides with formal guarantees.

The per-engine diagnostic profile routes each alert to the appropriate response: MCD-dominant alerts suggest measurement anomalies; Peña-dominant alerts suggest outbreak clusters; SDC-dominant alerts trigger immediate novel pathogen protocols. This aligns with priorities of the National Biosurveillance Strategy (HHS, 2022) and CDC BioSense Platform requirements for interpretable, real-time mixed-data anomaly detection.

9 Limitations

We document three limitations that bound the scope of the current results and suggest directions for future work.

Computational complexity. The dominant computational cost of MODA*-G is the pairwise Gower-MAD distance matrix, computed in $O(n^2(p+q))$ time and $O(n^2)$ space. For datasets with $n > 10{,}000$, this becomes prohibitive without approximation. Approximate kNN methods—including ball trees, KD-trees adapted for mixed data, and random projection approximations—can reduce this to $O(n \log n)$ without substantially affecting detection accuracy. Implementation of these approximations is deferred to future work.

Theoretical scope under (C3′). Condition (C3′) extends the standard observability assumption to allow categories absent from the clean distribution P_0 by working under the mixture model $P = (1-\varepsilon)P_0 + \varepsilon P_{\text{out}}$. The formal consistency results (Theorem 1 and Proposition 2) apply when $\varepsilon > 0$ so that all categories have positive probability under P. At the boundary $\varepsilon \to 0$, the unseen-category branch of SDC operates via the Laplace-smoothed estimator $\hat{P}_{\text{unseen}} = 1/(n + m_k + 1) \to 0$, which is not covered by the SLLN convergence proof. A complete treatment for the $\varepsilon \to 0$ boundary case—potentially using Good-Turing smoothing (Good, 1953) or support-based formulations—is a direction for future theoretical work. Empirically, SDC achieves AUC = 1.000 for O4 at $\varepsilon \leq 0.05$, confirming practical reliability despite this boundary.

Asymptotic rate without explicit variance. Proposition 2 establishes the $O_P(n^{-1/2})$ rate and asymptotic normality of MODA*-G$_n(x^*; k)$, but does not provide a closed-form expression for the asymptotic variance $\sigma^2_{\text{MODA}^*\text{-G}}(x^*; k)$. This quantity involves the joint asymptotic covariance matrix Σ of the four engine scores, which depends on the unknown data-generating distribution P in a complex way. Estimation of $\sigma^2_{\text{MODA}^*\text{-G}}$ and construction of confidence intervals for the MODA*-G score are deferred to future work. Practitioners requiring uncertainty quantification can use bootstrap resampling as an interim approach.

10 Conclusion

We have presented MODA*-G, a unified anomaly detection framework for mixed numerical and categorical data. Five formal theorems establish consistency, 50% breakdown point, partial affine equivariance, score comparability, and stochastic dominance. Two convergence propositions establish that the Softmax Gating weight of the dominant engine converges to one and that the MODA*-G score converges at rate $O_P(n^{-1/2})$. Monte Carlo simulation confirms that MODA*-G with $k = 3$

achieves mean AUC-ROC of 0.978, outperforming Isolation Forest in all four canonical contamination scenarios simultaneously. On the Annthyroid real-data benchmark (Campos et al., 2016), MODA*-G achieves AUC = 0.939, surpassing all six competing methods including EIF, COPOD, CatBoost-AD, and LOF, and exceeding the best result of Campos et al. (2016) by +0.189 AUC points.

The central insight is that the failure of linear combinations—demonstrated across five iterative implementation versions—is structural: fixed weights cannot adapt when an engine actively inverts rankings. Softmax Gating resolves this with three lines of code and no heuristic rules, yielding an attention mechanism that automatically amplifies the correct detector at the observation level. The real-data benchmark confirms that the advantage of native mixed-data operation is not merely theoretical: encoding categorical features as integers systematically degrades all competing methods on datasets with substantial categorical content.

Acknowledgements

This work was conducted independently. The author thanks the open-source communities behind NumPy, SciPy, and scikit-learn, which supported the simulation study.

References

Campos, G. O., Zimek, A., Sander, J., Campello, R. J. G. B., Micenková, B., Schubert, E., Assent, I., and Houle, M. E. (2016). On the evaluation of unsupervised outlier detection: measures, datasets, and an empirical study. *Data Mining and Knowledge Discovery*, 30(4):891–927.

Hariri, S., Kind, M. C., and Brunner, R. J. (2019). Extended isolation forest. *IEEE Transactions on Knowledge and Data Engineering*, 33(4):1479–1489.

Li, Z., Zhao, Y., Botta, N., Ionescu, C., and Hu, X. (2020). COPOD: copula-based outlier detection. *IEEE International Conference on Data Mining (ICDM)*, pp. 1118–1123.

Malinin, A., Chien, J., Gales, M., Knill, K., Kumar, S., Rayner, P., Shafran, I., and Williams, B. (2020). Uncertainty estimation in autoregressive structured prediction. *arXiv preprint arXiv:2002.07650*.

Bahdanau, D., Cho, K., and Bengio, Y. (2015). Neural machine translation by jointly learning to align and translate. *International Conference on Learning Representations (ICLR 2015)*.

Breunig, M. M., Kriegel, H.-P., Ng, R. T., and Sander, J. (2000). LOF: identifying density-based local outliers. *ACM SIGMOD Record*, 29(2):93–104.

Cabral, E. F., Vinces, B. V. S., Silva, G. D. F., Sander, J., and Cordeiro, R. L. F. (2025). Efficient outlier detection in numerical and categorical data. *Data Mining and Knowledge Discovery*, 39(3):1–46.

Devroye, L., Györfi, L., and Lugosi, G. (1996). *A Probabilistic Theory of Pattern Recognition*. Springer, New York.

Fawcett, T. (2006). An introduction to ROC analysis. *Pattern Recognition Letters*, 27(8):861–874.

Good, I. J. (1953). The population frequencies of species and the estimation of population parameters. *Biometrika*, 40(3-4):237–264.

Gower, J. C. (1971). A general coefficient of similarity and some of its properties. *Biometrics*, 27(4):857–871.

U.S. Department of Health and Human Services (2022). *National Biosurveillance Strategy 2022–2026*. Washington, D.C.

Jordan, M. I. and Jacobs, R. A. (1994). Hierarchical mixtures of experts and the EM algorithm. *Neural Computation*, 6(2):181–214.

Liu, F. T., Ting, K. M., and Zhou, Z.-H. (2008). Isolation forest. *Proceedings of the 8th IEEE International Conference on Data Mining*, pp. 413–422.

Marcano, L. and Fermín, W. (2013). Comparación de métodos de detección de datos anómalos multivariantes mediante un estudio de simulación. *Saber*, 25(2):192–201. Universidad de Oriente, Venezuela.

Peña, D. and Prieto, F. J. (2001). Multivariate outlier detection and robust covariance matrix estimation. *Technometrics*, 43(3):286–310.

Rousseeuw, P. J. (1985). Multivariate estimation with high breakdown point. *Mathematical Statistics and Applications*, 8:283–297.

Rousseeuw, P. J. and Van Driessen, K. (1999). A fast algorithm for the minimum covariance determinant estimator. *Technometrics*, 41(3):212–223.

Rousseeuw, P. J. and Hubert, M. (2018). Anomaly detection by robust statistics. *WIREs Data Mining and Knowledge Discovery*, 8(2):e1236.

Rousseeuw, P. J. and Van Zomeren, B. C. (1990). Unmasking multivariate outliers and leverage points. *Journal of the American Statistical Association*, 85(411):633–651.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., and Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems (NeurIPS)*, 30.

Wang, Y. and Li, Y. (2021). Outlier detection based on weighted neighbourhood information network for mixed-valued datasets. *Information Sciences*, 564:396–415.

Yuan, Z. et al. (2023). Attribute-weighted outlier detection for mixed data based on parallel mutual information. *Expert Systems with Applications*.

A Development History

MODA*-G emerged from a systematic iterative development process. The key insight—that fixed linear combinations are structurally incapable of handling active ranking inversion—required six implementation versions to establish empirically.

v0.1–v0.2 Four-engine framework with fixed weights. SDC implementation bug (hardcoded category 4 colliding with clean data) caused O4 AUC = 0.341. Fix: guaranteed-absent code (99) with proper Laplace smoothing restored AUC = 1.000 for SDC alone.

v0.3 Adaptive base weights $(0.30, 0.20, 0.15, 0.35)$ and Gower local+global distance combination. Results: O1: 0.951, O2: 0.983, O3: 0.936, O4: 0.513. O4 fails because Gower (γ=0.15) and MCD (α=0.30) actively invert rankings for compact categorical clusters.

v0.4–v0.4c Three variants of coherence weighting using Spearman correlations and separation criteria. Each variant improved some scenarios while degrading others. Root cause: all approaches relied on the uniform-weighted score to identify candidate outliers, which fails when the uniform score is itself dominated by inverting engines.

v0.5 Rank-based KS informativeness weighting. Failed across all scenarios: converting scores to ranks compresses information at the extremes. With $\varepsilon = 0.01$ (3 outliers in 300), the rank difference between the outlier (rank 300) and the nearest normal (rank 299) is $1/300 = 0.003$, while the raw score difference was 0.85. Ranks discard the most relevant signal.

v0.6 = MODA*-G Softmax Gating. Three lines of code, no heuristics, no thresholds. Mean AUC = 0.978 across all four scenarios.